CpE 3110 Homework Assignment #1 Maximum – 100pts

You must always show or explain your work in a **neat and orderly format**. You are encouraged to discuss ideas with other students and consult references, but your work must be your own.

Be sure to **show your work/derivation** for full credit (or maximum possible partial credit, if your final answer is not right). Also, do your best to maximize your HW score, since 13% of your semester letter grade will be decided by your HW performance.

1. Calculate the clock cycle frequency for 0.5ns clock cycle time. If the clock cycle time gets reduced by 20%, what will the resulting clock frequency be? [20pts]

2. Calculate the CPU time to execute a program containing one million machine instructions, when the average CPI of the CPU is 2.5 and CPU clock frequency is 2GHz. [20pts]

3. Let us overclock the machine given in problem 2 by increasing the CPU clock frequency from 2GHz to 2.5GHz. Calculate the CPU time of the overclocked machine running the same program containing one million instructions. How much faster (i.e., performance ratio) is the overclocked machine than the original machine? [20pts]

4. Consider two CPUs based on the same ISA (Instruction Set Architecture), namely CPU A and CPU B. CPU A's average CPI is 2.0 and CPU B's average CPI is 3.0. CPU B's clock frequency is also given as 2GHz. If CPU A is two times faster than CPU B in executing the same machine instruction sequence, what is CPU A's clock frequency? [20pts]

5. Suppose that we are considering an enhancement that runs 10 times faster than the original machine by is usable only 50% of the time. What is the overall speedup (i.e., performance ratio) gained by incorporating the enhancement? [20pts]